
【Hengyu Zhou】 Quick question on guideline refinement direction22 封邮件

Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

2026年5月11日 16:47

收件人: Kim Konwoo <ken1204@nii.ac.jp>

Hi Ken-san,

I presented your paper at our lab seminar today, it sparked a great discussion, especially around the open question of intrinsic guideline evaluation.

Professor Goto suggested I reach out: are you currently working on anything in that direction (e.g., evaluation criteria for guideline itself 's quality beyond downstream F1)? If so, I'd love to hear more about it, and happy to help in any way I can.

Thanks!
Hengyu Zhou

Kim Konwoo <ken1204@nii.ac.jp>

2026年5月12日 11:25

收件人: Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

Hi Hengyu,

Yes, actually I am currently working on guideline generation and evaluation. For guideline generation, I am utilizing existing resources, such as schema and gold annotations, to develop initial guidelines and use my ACL approach to develop further. For evaluation, I am still considering how I should approach it, especially intrinsic evaluation. For extrinsic evaluation, I may use gold annotations to kind of measure how much refined guidelines could help, but it is still unclear how we could measure the quality of refined guidelines. Have you made any progress related to your research? I would also love to hear anything for inspiration.

Best,

Kon Woo Kim
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Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

2026年5月14日 16:19

收件人: Kim Konwoo <ken1204@nii.ac.jp>

Hi Ken-san,

Thanks for the update. I think I have a clear understanding of the extrinsic evaluation methodology from your previous paper.

As I'm still refining my research theme, I want to dive deeper into the intrinsic evaluation of guidelines. If you have any recent papers on your reading list related to this area, would you mind sharing them with me?

Best wishes,
Hengyu Zhou

Kim Konwoo <ken1204@nii.ac.jp> 于2026年5月12日周二 11:25写道:
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Kim Konwoo <ken1204@nii.ac.jp>

2026年5月16日 20:19

收件人: Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

Hi Hengyu-san,

I am sorry to inform you that I have not read much recent papers regarding intrinsic evaluation. I am currently writing a draft for another conference, and I am focusing on the results derived from our own datasets. In the manuscript, I also stated that we leave intrinsic evaluation and optimization as future work. I think that will be my main focus in the next paper.

Ken

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Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

2026年5月19日 08:10

收件人: Kim Konwoo <ken1204@nii.ac.jp>

Hi Ken-san,

Thank you for the update!

It's great to know that intrinsic evaluation will be the focus of your next paper. I'm also moving in that direction, so perhaps we can exchange ideas once things settle down on your end.

Good luck with the conference submission!

Best,

Hengyu Zhou

Kim Konwoo <ken1204@nii.ac.jp> 于2026年5月16日周六 20:19写道:

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Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

2026年5月26日 12:14

收件人: Kim Konwoo <ken1204@nii.ac.jp>

Hi Ken-san,

Thank you again for sharing your paper, I've been reading it carefully again and had a couple of questions I was hoping you could help clarify.

- Regarding the 10-document dev set: could you share more about how the documents were selected and the reasoning behind choosing that particular size? I'm also curious about your thoughts on how scaling up the dev set might affect the moderation process.
- Could you provide more details on the prompting strategies used in the discrepancy analysis (confusion matrix), pattern explanation, and principle generation steps? The paper gives a clear overview of the structure, but I'd love to understand the specifics of how the prompts were designed.

I really appreciate any insight you can share!

Best,

Hengyu Zhou

Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp> 于2026年5月19日周二 08:10写道:

[引用文字已隐藏]

Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

2026年6月2日 13:58

收件人: Kim Konwoo <ken1204@nii.ac.jp>

Hi Ken-san,

Hope things are going well! I wanted to gently follow up on my questions from May 26th regarding the dev set selection and prompting strategies, no rush at all if you're still deep in the submission crunch.

I'm also looking forward to hearing your thoughts when you have a moment.

Best,

Hengyu

2026年5月26日(火) 12:14 Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>:

[引用文字已隐藏]

Kim Konwoo <ken1204@nii.ac.jp>

2026年6月8日 00:45

收件人: Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

Hi Hengyu-san,

Sorry for the late reply. I was back in Korea for a family issue and just got back now.

To answer your question,

(1) Regarding the 10-document dev set, 10 documents were randomly selected from TRAIN set in each dataset. The reason we chose 10 documents was more heuristics. In many real annotation projects, human annotators first annotate small sets (numbers vary depending on how big their projects are) and start moderation with those sets to resolve misalignments. In our paper, we only had 100 documents for evaluation, so we only used 10 documents. Another reason we used 10 documents is that we did not have enough time and budget to run more documents, such as 20 or 30, for moderation. If enough time and budget were given, we could definitely have tried probably more.

(2) If you refer to this repo, you should be able to look at more details regarding prompting: <https://github.com/KonWooKim/llm-guideline-moderation>. Prompting itself is not very special, but I think the logic is more important. Also, the prompting methods we used for the ACL paper needs to be improved. Our moderation framework consists of three stages, pattern inference, principle generation, and guideline refinement. The first stage focuses on finding a common convention based on the biggest error group; the second stage focuses on generating a principle that can cover those discrepancies; and the last stage is used to update the original guidelines.

Ken

2026년 5월 26일 (화) 오후 12:14, Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>님이 작성:

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Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

2026年6月11日 12:52

收件人: Kim Konwoo <ken1204@nii.ac.jp>

Hi Ken-san,

Thank you so much for sharing the repository, it has been incredibly helpful!

I am currently preparing to replicate your work on the NCBI Disease dataset. As I have been thinking through the framework, I have two ideas to reduce potential side effects and would like to hear your thoughts:

(1) Conflict check before principle integration: After generating a new principle in principle generation, I am considering adding a step that checks it against existing guidelines for logical contradictions or coverage overlap (not only the in prompt TP check) before guideline refinement. Do you think this is worthwhile, or is the current refinement stage already robust to this?

(2) Top-k error pattern modification: Rather than targeting only the single largest error group in pattern inference, I am considering simultaneously addressing the top-k patterns from the confusion matrix. Do you think this is a practical direction?

I would greatly appreciate any feedback based on your experience.

Best regards,
Hengyu Zhou

2026年6月8日(月) 0:45 Kim Konwoo <ken1204@nii.ac.jp>:

[引用文字已隱藏]

Kim Konwoo <ken1204@nii.ac.jp>

2026年6月19日 00:38

收件人: Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

Hi Hengyu-san,

I am sorry about the late reply. I just had my intermediate presentation this week, so I needed some time to get prepared.

To answer your questions,

(1) Yes, I think we need to do some conflict checks either before or after the refinement. I think that will be very worthwhile.

(2) To answer for top-k errors, I will attach another paper that I submitted to EMNLP this year. It is still under review, so please do not share with others yet (except Goto-sensei). If you read this paper, I tried a top-k case card approach to address multiple errors at the same time. It helps improve performance, but you will also find that (1) was actually needed to confirm whether we could monotonically improve guidelines and model performance because latter case cards actually conflicted with previous cards and led to performance drop (It is written in results and discussion sections).

Please let me know if you have any questions.

Best regards,

Kon Woo Kim
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 **EMNLP2026.pdf**
313K

Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>
收件人: Kim Konwoo <ken1204@nii.ac.jp>

2026年6月23日 17:52

Hi Ken-san,

Thanks for the detailed reply, and congratulations on getting through your intermediate presentation. No worries about the timing at all. I've also been slow this week with a lot of interviews on my end.

Thanks as well for sharing the EMNLP submission. I'll keep it confidential and only discuss it with Goto-sensei. I'm looking forward to reading it carefully, especially the top-k case card approach for handling multiple errors at once. Your point about how the conflict checks in (1) were needed to confirm monotonic improvement is very interesting. This is exactly the kind of issue I want to understand better

On my end, I've reproduced your experimental results on the NCBI Disease dataset using GPT-5.4. After that, I'd like to try the first improvement direction (the conflict checks).

I'll follow up once I've gone through the paper. Let me know if there's anything in particular you'd like me to focus on.

Best regards,
Hengyu

2026年6月19日(金) 0:38 Kim Konwoo <ken1204@nii.ac.jp>:
[引用文字已隐藏]

Kim Konwoo <ken1204@nii.ac.jp>
收件人: Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

2026年6月26日 14:11

Hi Hengyu-san,

I am actually curious about your replicated results. Could you please provide any updates once you finish it?

Also, which aspect are you particularly interested in? Are you still interested in applying RAG-based approaches to guidelines? It would be helpful if you could tell me what ideas or directions you have so far.

Ken
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Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>
收件人: Kim Konwoo <ken1204@nii.ac.jp>

2026年6月26日 15:59

Hi Ken-san,

Thanks for your interest on the reproduction result, I attached the result report in the attachment. This is the result of moderation, and as for the validation part, I plan to first reproduce the top-k case card moderation, then test the performance on validation dataset.

As for the current direction, I don't have a clear idea. I was planning to apply the top-k case card moderation, now I'm considering how to avoid the side effects. I will first read your most recent paper then try to think some new ideas.

As for the RAG-based approaches, I'm considering where to apply RAG? 1. The LLM has enough knowledge to make annotation; 2. The context window is large enough to handle the guidelines. I appreciate if you can give any insights on this direction.

Best regards,
Hengyu

Kim Konwoo <ken1204@nii.ac.jp> 于2026年6月26日周五 14:11写道:
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 **reproduction_report.pdf**
248K

Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>
收件人: Kim Konwoo <ken1204@nii.ac.jp>

2026年7月14日 08:48

Hi Ken-san,

Thank you again for your continued guidance, it has been extremely helpful for my reproduction work.

I would like to ask: beyond the top-k case card moderation and intrinsic guideline refinement directions we've discussed, what do you see as your own next research focus in this line of work?

Separately, I'm now starting to think about the topic for my own master's thesis in this area. If you have any suggestions on possible themes, or directions that might complement (rather than overlap with) your ongoing work, I would be extremely grateful for your advice.

Thank you very much for your time.

Best regards,
Hengyu

2026年6月26日(金) 15:59 Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>:
[引用文字已隐藏]

Kim Konwoo <ken1204@nii.ac.jp>
收件人: Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

2026年7月16日 02:48

Hi Hengyu-san,

Thank you for the email.

Before I make any suggestions, could you please tell me a bit more about your current status, research interests, and strengths (for example, whether you are more experienced with deep learning, statistical approaches, and etc.)?

I would also like to learn more about your research goals, what Goto-sensei has suggested so far, and what you hope to do after graduation.

Please feel free to share as much detail as you can.

Ken
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Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>
收件人: Kim Konwoo <ken1204@nii.ac.jp>

2026年7月16日 11:35

Hi Ken-san,

Thank you very much for your reply and for offering to help.
Here is some more detail about my current situation:

1. Current status:

I'm still at the topic-selection stage. So far I've only reproduced your prior study related to guideline refinement, and I haven't started any experiments beyond that yet.

2. Research interests / strengths:

I don't have much hands-on research experience yet in either deep learning or statistical approaches, but statistics is a relative weak point for me.

I find myself more drawn to the engineering side of research. I'm particularly interested in questions around efficiency, scalability, and system-level implementation, and I tend to care more about how a method performs in real, practical scenarios than about theoretical novelty for its own sake.

3. Research goal:

My only concrete direction so far is guideline refinement, but since I've only reproduced your paper, I don't yet have a clear sense of where to take it from here. I'm also open to other topics!

4. Goto-sensei's suggestions so far: I'm meeting with Goto-sensei tomorrow to clarify this, including the expected scope/workload for a master's thesis. I'll follow up with the details soon.

5. After graduation: I'm hoping to move toward an engineering-focused career, most likely as a Machine Learning Engineer (MLE), rather than staying in a research-heavy role.

Please let me know if there's anything else that would help you give suggestions. Happy to share more detail!

Thanks again for your help,
Hengyu

Kim Konwoo <ken1204@nii.ac.jp> 于2026年7月16日周四 02:48写道:
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Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

2026年7月17日 21:27

收件人: Kim Konwoo <ken1204@nii.ac.jp>

Hi Ken-san,

Thank you for your patience. I met with Goto-sensei and would like to share what we discussed, plus my current take on the RAG idea you asked about.

Proposed direction:

The main idea is to improve your guideline refinement method and apply it to Eric (Zhao-san)'s work, where the original guideline was given directly to the LLM with prompt engineering. Since the NER datasets overlap between your paper and Eric's, I'm considering either other datasets, or extending to the Relation Extraction and multi-label Text Classification tasks, to avoid repeating existing experiments. I'd appreciate your thoughts on this.

On the RAG idea:

My current conclusion is that a retrieval-based approach isn't worth pursuing in this setting. The rule set is only ~21 patches (a few thousand tokens), so retrieval adds little value. It also doesn't resolve conflicts, since contradictory rules get retrieved together and the resolution still falls to the LLM. Instead, I think resolving conflicts before annotation is more promising: cluster related rules and merge contradictory ones into conditional meta-rules, producing one self-consistent guideline, then re-test whether F1 stays monotonic as more patches are added. If it does, that would show the method raises the ceiling on usable guideline size. I'd like to hear whether you think this holds up.

Two questions:

1. Do you happen to have any sub-project that could work as a topic for my master's thesis? I'd be glad to contribute wherever helpful.
2. Would you be available to join my meeting with Goto-sensei on the 27th? If so, I'll also invite Professor Kim.

Thanks again, and looking forward to your thoughts.

Best regards,
Hengyu

Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp> 于2026年7月16日周四 11:35写道:
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Kim Konwoo <ken1204@nii.ac.jp>

2026年7月21日 01:18

收件人: Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

Dear Hengyu-san,

I think you can try improving my method and applying it to Eric's work. If you are still planning to use RAG, I would suggest you find other datasets that have longer guidelines.

I have decided to split the current EMNLP draft into two separate papers because I do not think that 50 manually annotated samples are sufficient to support strong conclusions for a long paper at a top-tier conference. As a result, I am currently focusing on two directions:

- (1) Using the manual annotations from the past year, I plan to prepare and submit a journal paper.
- (2) I will also work on guideline generation. However, I will determine the details after I finish (1).

Therefore, I may abandon the draft that I shared with you earlier (the multi-annotator setting). If you are interested, you are welcome to extend that work. However, there are several prerequisites that should be addressed before the idea can be meaningfully extended.

You may also want to take a look at this paper, which follows a similar approach to the one I proposed in the earlier draft:

<https://arxiv.org/abs/2604.15866>

To answer your questions,

1. Do you happen to have any sub-project that could work as a topic for my master's thesis? I'd be glad to contribute wherever helpful.

As of now, I think the disagreement-guided guideline improvement project based on the multi-annotator setting would be a suitable sub-project.

2. Would you be available to join my meeting with Goto-sensei on the 27th? If so, I'll also invite Professor Kim.

Yes, I am available to join the meeting on the 27th.

Best,

Kon Woo Kim

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Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

2026年7月22日 16:41

收件人: Kim Konwoo <ken1204@nii.ac.jp>

Hi Ken-san,

Thank you for the detailed reply.
I'll follow up on the other points separately.

Thank you also for agreeing to join the meeting on the 27th, I really appreciate it!

Would 15:00 (Tokyo time) work for you? If another time is better, I can adjust it with Goto-sensei.

Once the time is confirmed, I'll send out the meeting link.

Best regards,
Hengyu

2026年7月21日(火) 1:19 Kim Konwoo <ken1204@nii.ac.jp>:
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Kim Konwoo <ken1204@nii.ac.jp>

2026年7月22日 17:00

收件人: Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

Yes, 15:00 on 7/27 works for me.

Ken

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Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

2026年7月23日 13:09

收件人: Kim Konwoo <ken1204@nii.ac.jp>

Hi ken-san,

Thank you for confirming. Here is the link for the meeting on July 27 at 15:00 (Tokyo time):

<https://u-tokyo-ac-jp.zoom.us/j/3344201271>

Professor Kim will be joining as well.

See you then.

Best regards,
Hengyu

2026年7月22日(水) 17:00 Kim Konwoo <ken1204@nii.ac.jp>:

[引用文字已隐藏]

Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp>

2026年7月27日 18:19

收件人: Kim Konwoo <ken1204@nii.ac.jp>

Hi Ken-san,

Thank you very much for joining today's meeting and for the detailed discussion.

The feedback was very helpful, especially the suggestion to build on the moderation framework from the ACL paper rather than starting from a new one.

As a next step, I will first reproduce the ACL paper's framework on NCBI Disease, BC5CDR, and BioRED. Beyond the reported F1, I plan to record the number of iterations, the guideline length at each iteration, and the variance across repeated runs, since these will determine which direction is worth pursuing. I will share the results with you once they are ready.

Thank you again for your time.

Best regards,
Hengyu

Hengyu ZHOU <zhou-hengyu@g.ecc.u-tokyo.ac.jp> 于2026年7月23日周四 12:09写道:

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